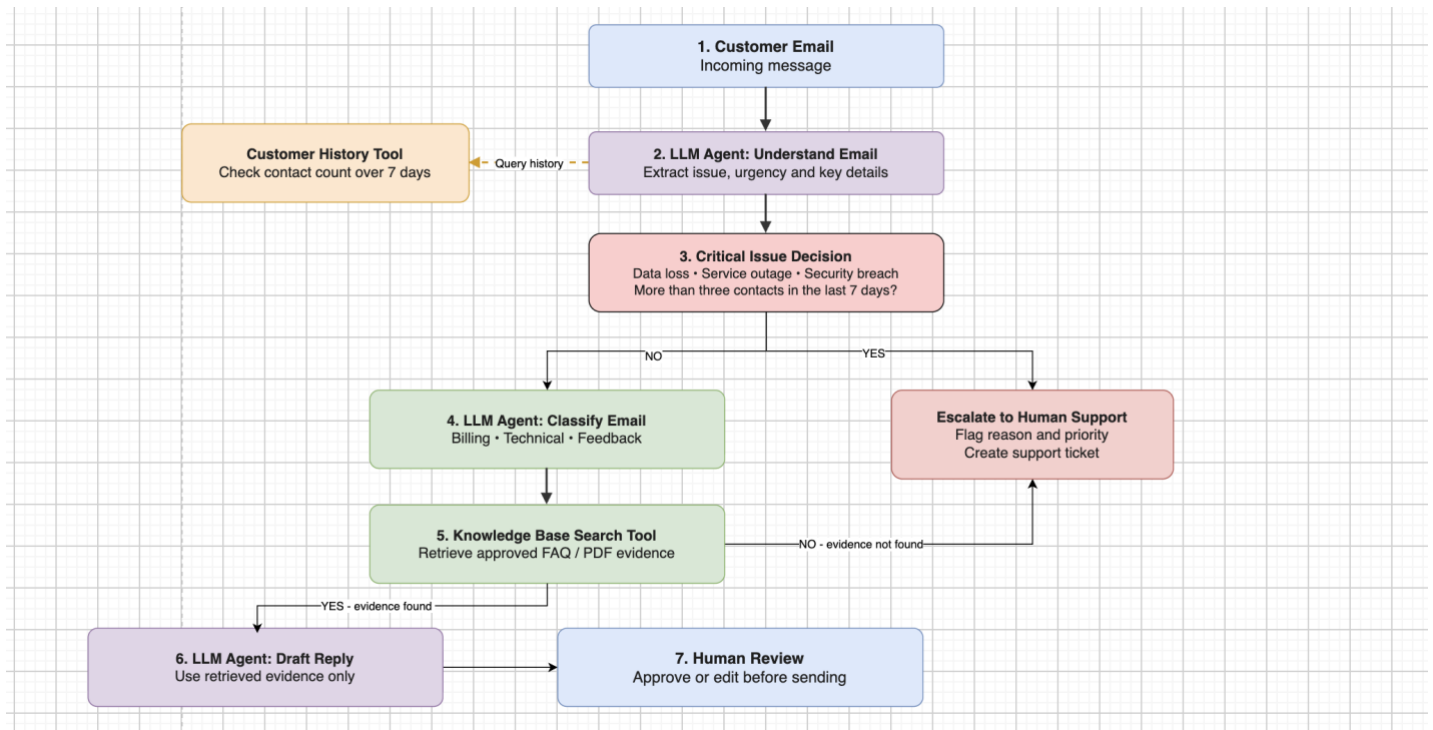


AGENTIC ARCHITECT CHALLENGE



1. System Architecture

The system uses an LLM agent to help process customer support emails. When an email arrives, the LLM reads the message and identifies the main issue. The system then checks whether the email involves a critical issue, such as data loss, service outages or security breaches. It also checks whether the customer has contacted support more than three times in the past seven days. If any of these conditions are met, the email is sent to a human support agent. Otherwise, the LLM classifies the email into Billing, Technical or Feedback.

Next, the Knowledge Base Search Tool searches the internal FAQs and PDFs for relevant information. The LLM uses this information to draft a reply instead of making up its own answers. If no suitable information is found, the case is sent to a human. Finally, a human reviews the reply before it is sent to the customer.

2. Design Trade-offs

The system uses AI to understand emails, classify them and draft replies. However, critical issues are handled by human agents, and replies are checked before sending. This makes the system safer but takes more time. The system also uses internal FAQs and PDFs instead of relying only on the LLM's own knowledge. This helps reduce incorrect answers, especially about refund policies, but the answers depend on the information available in the knowledge base.

3. Potential Failure Points and Mitigations

The system checks the email and customer history to avoid missing critical issues. If an outage is unclear, it checks the service status. To prevent incorrect policy information, the system uses internal FAQs and PDFs. If no suitable answer is found, the case is sent to a human. If a tool fails or information is unavailable, the system also escalates the case to human support. Finally, emails are classified into Billing, Technical or Feedback, and a human checks the reply before it is sent.